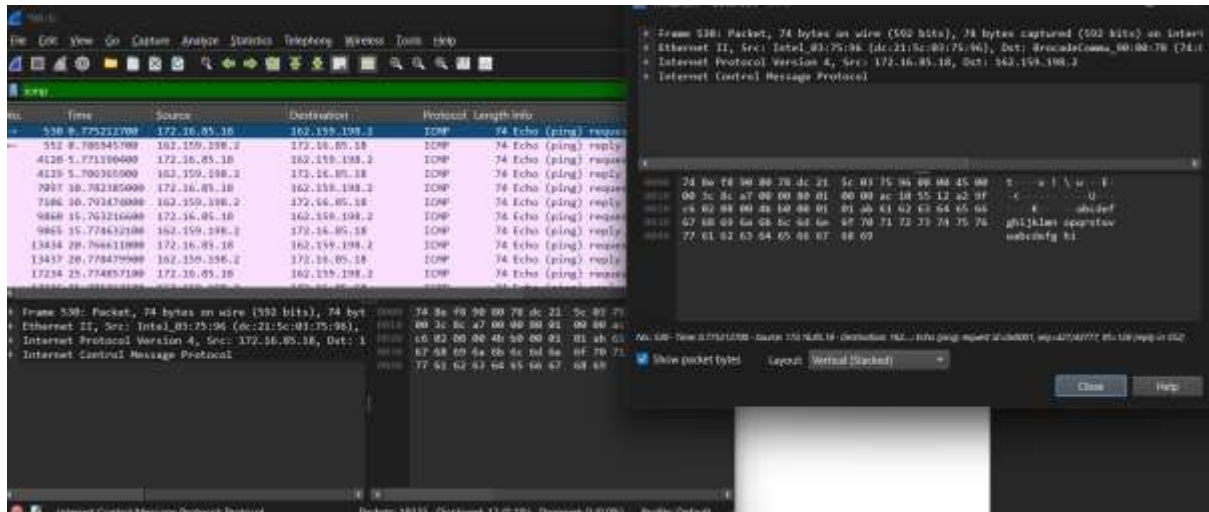


Advance Computer network PCS223

Assignment-2

Pranav Bhardwaj 8025320073

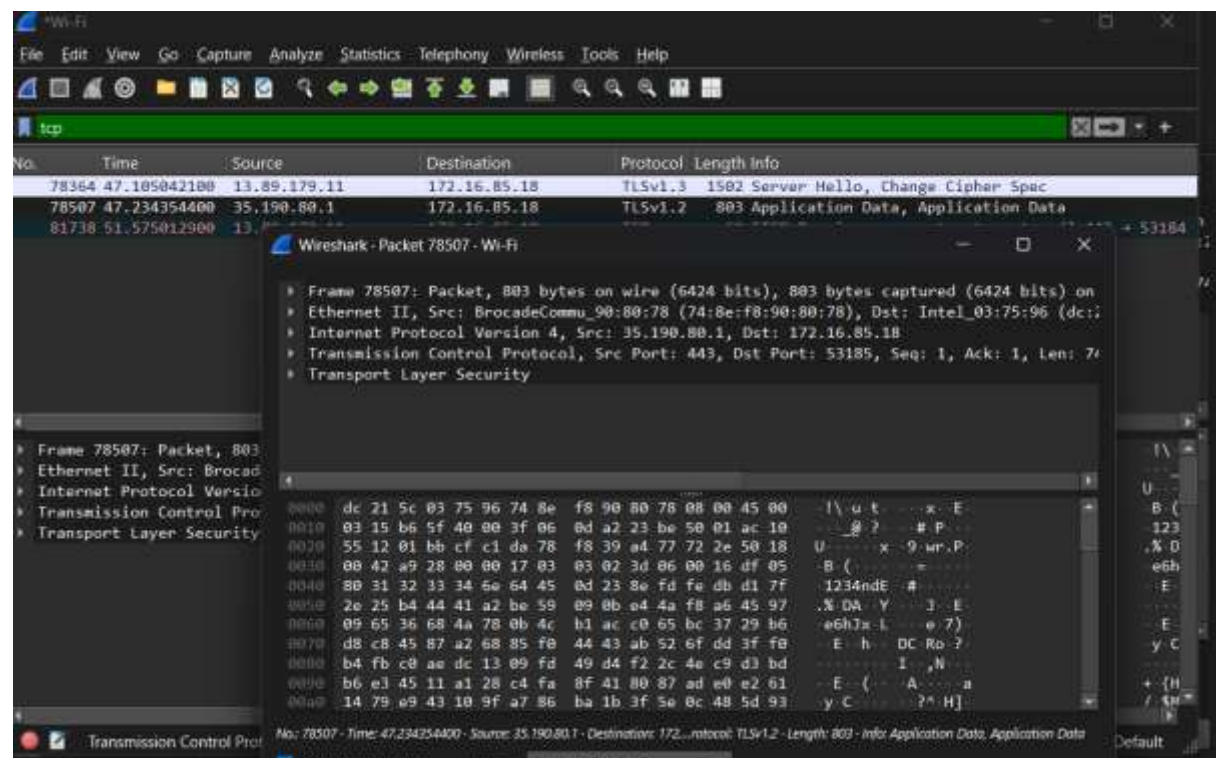
1. To capture and analyze ICMP packets (Ping request and reply)



Observation

- Interface used: Wi-Fi
- Protocol: ICMP
- Type: Echo (Ping) Request
- Source IP address: 172.16.85.18
- Destination IP address: 162.159.198.2
- Packet length: 74 bytes

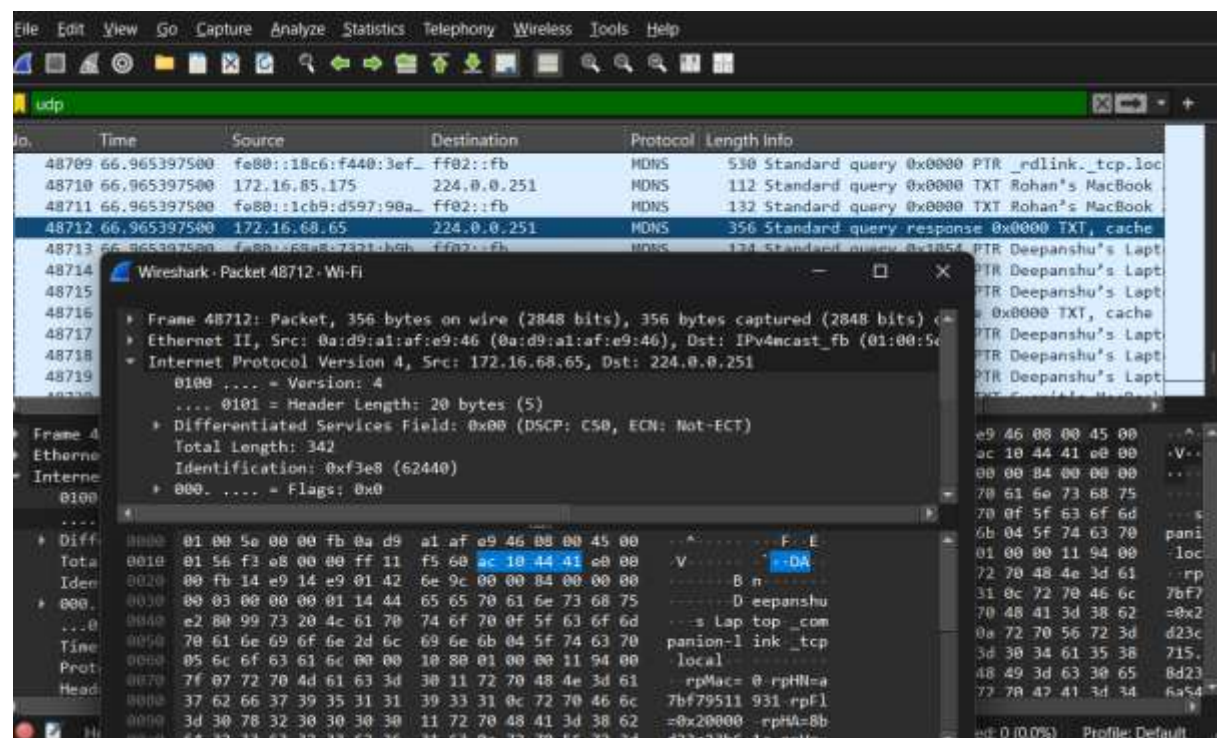
2. To analyze TCP packets and observe the TCP three-way handshake



Observation

- Interface used: Wi-Fi
- Protocol: TCP
- Source IP address: 13.89.179.11
- Destination IP address: 172.16.85.18
- Source Port: 443
- Destination Port: 53184
- Flags observed: FIN, ACK
- Packet length: 60 bytes

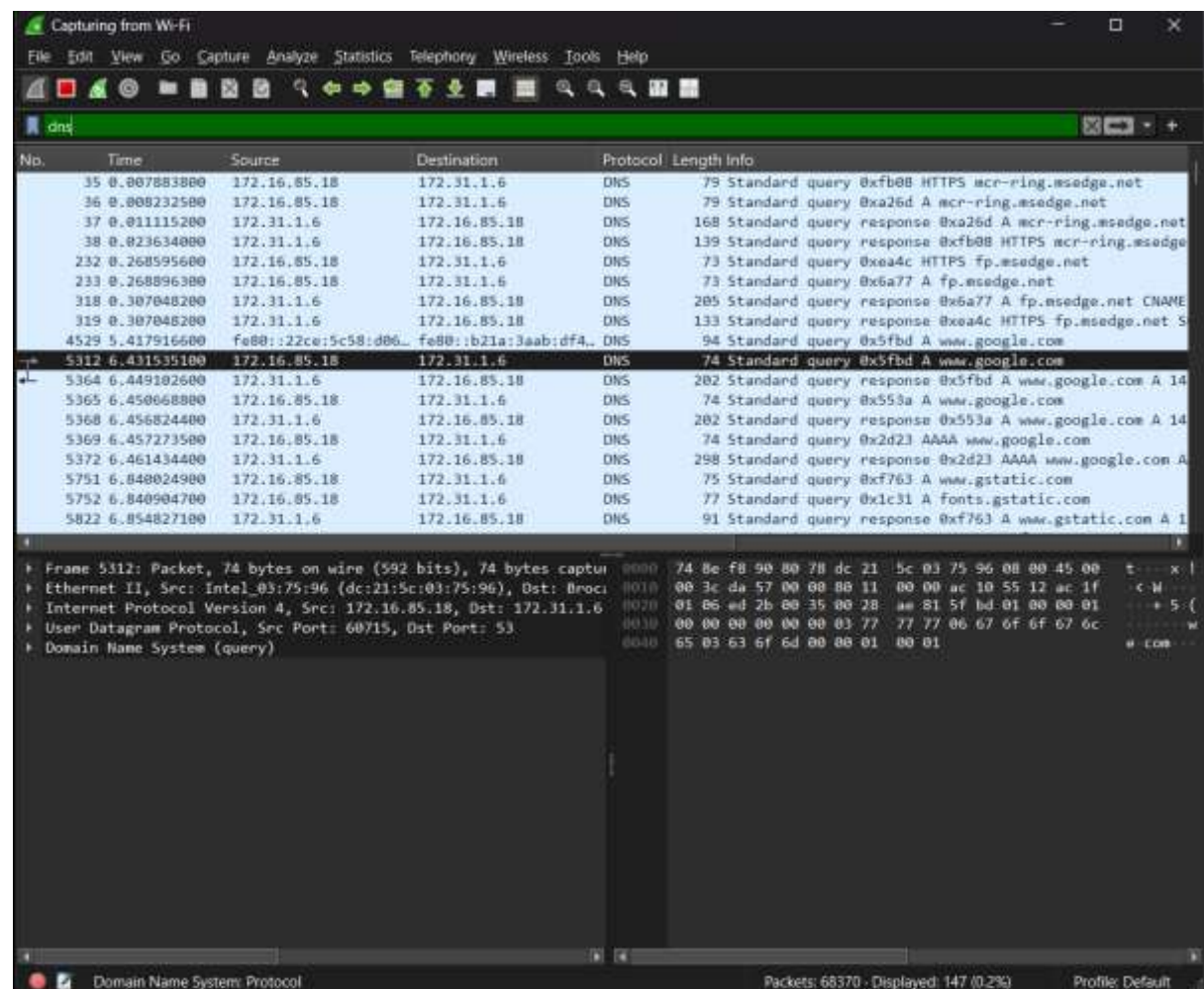
3. To capture and analyze UDP packets



Observation (UDP Experiment)

- Interface used: Wi-Fi
- Protocol: UDP
- Source IP address: 172.16.68.65
- Destination IP address: 224.0.0.251
- Source Port: 5353
- Destination Port: 5353
- Packet length: 356 bytes
- Application layer protocol: DNS (mDNS)

4. To capture and analyze DNS query and response packets



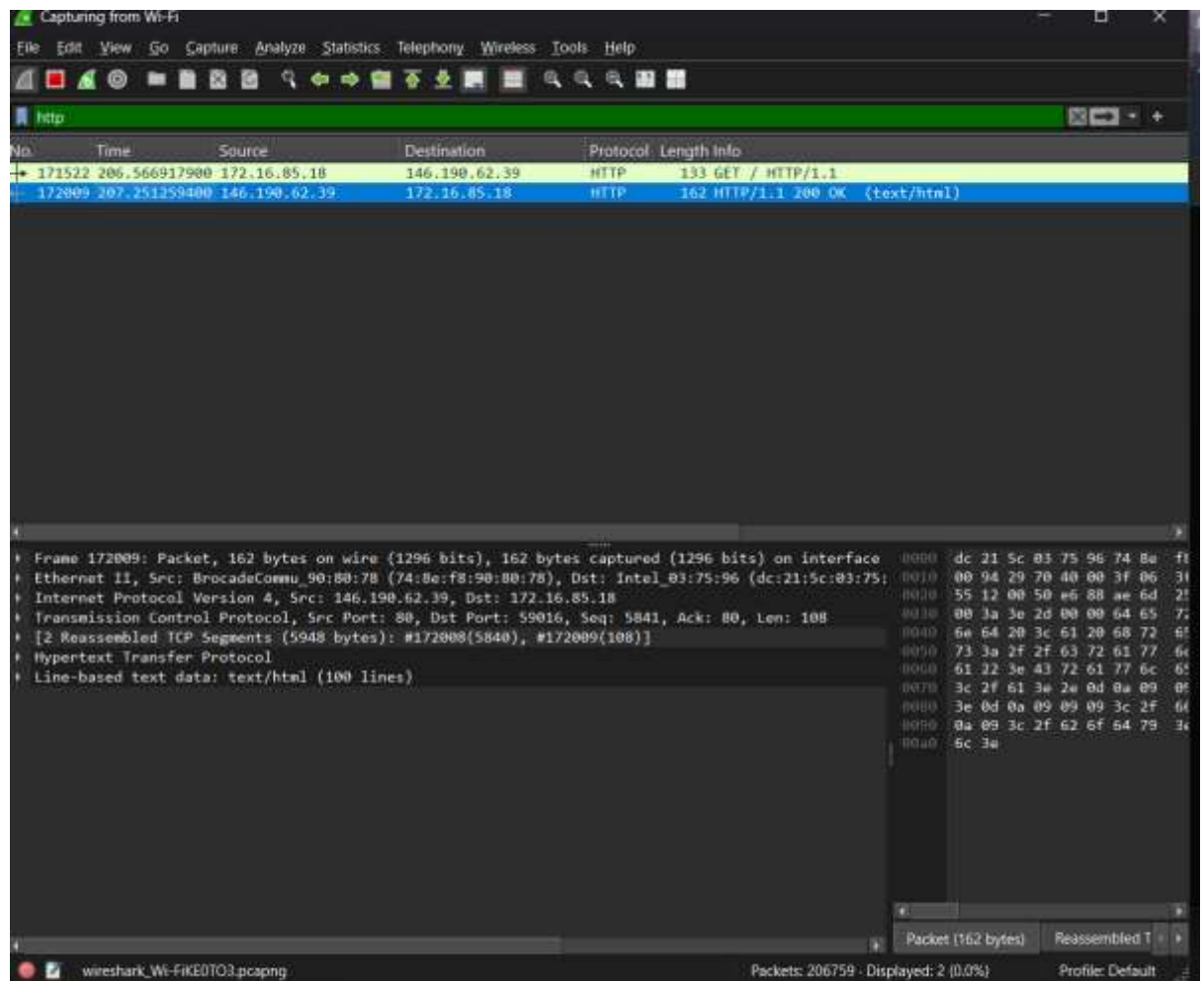
No.	Time	Source	Destination	Protocol	Length	Info
35	0.007883800	172.16.85.18	172.31.1.6	DNS	79	Standard query 0xfb08 HTTPS mcr-ring.msedge.net
36	0.008232500	172.16.85.18	172.31.1.6	DNS	79	Standard query 0xa26d A mcr-ring.msedge.net
37	0.011115200	172.31.1.6	172.16.85.18	DNS	168	Standard query response 0xa26d A mcr-ring.msedge.net
38	0.023634000	172.31.1.6	172.16.85.18	DNS	139	Standard query response 0xfb08 HTTPS mcr-ring.msedge.net
232	0.268595600	172.16.85.18	172.31.1.6	DNS	73	Standard query 0xea4c HTTPS fp.msedge.net
233	0.268896300	172.16.85.18	172.31.1.6	DNS	73	Standard query 0x6a77 A fp.msedge.net
318	0.307048200	172.31.1.6	172.16.85.18	DNS	205	Standard query response 0x6a77 A fp.msedge.net CNAME
319	0.307048200	172.31.1.6	172.16.85.18	DNS	133	Standard query response 0xea4c HTTPS fp.msedge.net S
4529	5.417916600	fe80::22ce:5c58:d86...	fe80::b21a:3aab:df4...	DNS	94	Standard query 0x5fbd A www.google.com
5312	6.431535100	172.16.85.18	172.31.1.6	DNS	74	Standard query 0x5fbd A www.google.com
5364	6.449102600	172.31.1.6	172.16.85.18	DNS	202	Standard query response 0x5fbd A www.google.com A 14
5365	6.450688800	172.16.85.18	172.31.1.6	DNS	74	Standard query 0x553a A www.google.com
5368	6.456824400	172.31.1.6	172.16.85.18	DNS	202	Standard query response 0x553a A www.google.com A 14
5369	6.457273500	172.16.85.18	172.31.1.6	DNS	74	Standard query 0x2d23 AAAA www.google.com
5372	6.461434400	172.31.1.6	172.16.85.18	DNS	298	Standard query response 0x2d23 AAAA www.google.com A
5751	6.848024900	172.16.85.18	172.31.1.6	DNS	75	Standard query 0xf763 A www.gstatic.com
5752	6.840904700	172.16.85.18	172.31.1.6	DNS	77	Standard query 0x1c31 A fonts.gstatic.com
5822	6.854827100	172.31.1.6	172.16.85.18	DNS	91	Standard query response 0xf763 A www.gstatic.com A 1

Frame 5312: Packet, 74 bytes on wire (592 bits), 74 bytes captured on interface Intel_03:75:96 (dc:21:5c:03:75:96), Dst: Brocade (00:0c:0c:00:00:00)
Ethernet II, Src: Intel_03:75:96 (dc:21:5c:03:75:96), Dst: Brocade (00:0c:0c:00:00:00)
Internet Protocol Version 4, Src: 172.16.85.18, Dst: 172.31.1.6
User Datagram Protocol, Src Port: 60715, Dst Port: 53
Domain Name System (query)

Observation

- Interface used: Wi-Fi
- Protocol: DNS
- Transport protocol: UDP
- Source IP address: 172.16.85.18
- Destination IP address: 172.31.1.6
- Source Port: 60715
- Destination Port: 53
- Query: www.google.com
- Packet length: 74 bytes

5. To capture and analyze HTTP packets



Observation

- Interface used: Wi-Fi
- Protocol: HTTP
- Transport protocol: TCP
- Source IP address: 172.16.85.18
- Destination IP address: 146.190.62.39
- Source Port: 59016
- Destination Port: 80
- Request Method: GET
- Response: HTTP/1.1 200 OK
- Content Type: text/html

The image shows a Wireshark packet capture titled "Capturing on Wi-Fi". The interface is set to "arp". The packet list shows multiple ARP requests from "BrocadeComm_90:80:10" to the broadcast address "172.16.64.1". Packet 348154 is selected, showing an ARP request for "172.16.77.115". The packet details pane shows the Ethernet II header and the ARP (Request) section. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
348123	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.72.227? Tell 172.16.64.1
348124	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.68.8? Tell 172.16.64.1
348125	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.73.116? Tell 172.16.64.1
348131	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.69.67? Tell 172.16.64.1
348136	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.86.202? Tell 172.16.64.1
348138	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.94.168? Tell 172.16.64.1
348139	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.81.64? Tell 172.16.64.1
348141	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.87.77? Tell 172.16.64.1
348142	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.79.39? Tell 172.16.64.1
348148	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.68.105? Tell 172.16.64.1
348149	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.94.58? Tell 172.16.64.1
348150	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.78.152? Tell 172.16.64.1
348154	426.486383500	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.77.115? Tell 172.16.64.1
348161	426.576448100	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.80.13? Tell 172.16.64.1
348202	426.587634700	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.94.109? Tell 172.16.64.1
348208	426.587634700	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.94.96? Tell 172.16.64.1
348211	426.587634700	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.85.161? Tell 172.16.64.1
348213	426.587634700	BrocadeComm_90:80:10	Broadcast	ARP	60	Who has 172.16.69.235? Tell 172.16.64.1

Frame 348154: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Dev Ethernet II, Src: BrocadeComm_90:80:7f (74:8e:f8:90:80:7f), Dst: Broadcast (ff:ff:ff:ff:ff:ff) Address Resolution Protocol (request)

0000 ff ff ff ff ff ff 74 8e ff
0010 08 00 06 04 00 01 74 8e ff
0020 00 00 00 00 00 00 ac 10 4c
0030 00 00 00 00 00 00 00 00 00

Address Resolution Protocol: Protocol

Packets: 351175 - Displayed: 22177 (6.3%) Profile: Default

- **Interface used: Wi-Fi**
- **Protocol: ARP**
- **Type: ARP Request**
- **Source MAC address: 74:8e:f8:90:80:78**
- **Destination MAC address: ff:ff:ff:ff:ff:ff (Broadcast)**
- **Query: Who has 172.16.77.115? Tell 172.16.64.1**
- **Packet length: 60 bytes**

7. To identify source and destination MAC and IP addresses in captured packets

8. Interface used: Wi-Fi

9. Source MAC address: 74:8e:f8:90:80:78

10. Destination MAC address: ff:ff:ff:ff:ff:ff

11. Source IP address: 172.16.64.1

12. Destination IP address: 172.16.77.115